CO2 – ASSESSMENT TOOL – 3

Question 1 – Breadth-First Search (BFS)

BFS visits nodes level by level, adding children from left to right.

Iteration	Queue before visit	Visited node	Queue after iteration
1	[A]	A	[B, C]
2	[B, C]	B	[C, D, E]
3	[C, D, E]	C	[D, E, F, G]
4	[D, E, F, G]	D	[E, F, G]
5	[E, F, G]	E	[F, G]
6	[F, G]	F	[G]
7	[G]	G	[]

1. BFS traversal order: A → B → C → D → E → F → G

2. Queue contents after each iteration: [B, C] → [C, D, E] → [D, E, F, G] → [E, F, G] → [F, G] → [G] → [].

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Question 2 – Depth-First Search (DFS)

DFS follows one branch as far as possible, then backtracks.

Step	Node visited	Explanation / active path
1	A	Start at A
2	B	Go to left child B
3	D	Go to left child D; leaf node
4	E	Backtrack to B, then visit E
5	C	Backtrack to A, then visit C
6	F	Go to left child F
7	G	Backtrack to C, then visit G

DFS traversal order: A → B → D → E → C → F → G

Explanation: Starting at A, DFS explores the left subtree B before moving to C. It visits D, backtracks to E, returns to A, and then explores C's children F and G.

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Question 3 – 8-Puzzle Using Breadth-First Search (BFS)

Initial state (0 is the blank tile)

1	3	6
5	0	2
4	7	8

Goal state

1	2	3
4	5	6
7	8	0

First five iterations: the queue is FIFO; newly generated states are added at the back.

Iteration	Current state expanded	Queue after expansion (new states added at back)
1	1 3 6 / 5 0 2 / 4 7 8	4 states; new: 1 0 6 / 5 3 2 / 4 7 8; 1 3 6 / 5 7 2 / 4 0 8; 1 3 6 / 0 5 2 / 4 7 8; 1 3 6 / 5 2 0 / 4 7 8
2	1 0 6 / 5 3 2 / 4 7 8	5 states; new: 0 1 6 / 5 3 2 / 4 7 8; 1 6 0 / 5 3 2 / 4 7 8
3	1 3 6 / 5 7 2 / 4 0 8	6 states; new: 1 3 6 / 5 7 2 / 0 4 8; 1 3 6 / 5 7 2 / 4 8 0
4	1 3 6 / 0 5 2 / 4 7 8	7 states; new: 0 3 6 / 1 5 2 / 4 7 8; 1 3 6 / 4 5 2 / 0 7 8
5	1 3 6 / 5 2 0 / 4 7 8	8 states; new: 1 3 0 / 5 2 6 / 4 7 8; 1 3 6 / 5 2 8 / 4 7 0

Shortest solution path (8 moves)

Step	Blank move	State (rows separated by /)
0	Initial	1 3 6 / 5 0 2 / 4 7 8
1	Right	1 3 6 / 5 2 0 / 4 7 8
2	Up	1 3 0 / 5 2 6 / 4 7 8
3	Left	1 0 3 / 5 2 6 / 4 7 8
4	Down	1 2 3 / 5 0 6 / 4 7 8
5	Left	1 2 3 / 0 5 6 / 4 7 8
6	Down	1 2 3 / 4 5 6 / 0 7 8
7	Right	1 2 3 / 4 5 6 / 7 0 8
8	Right	1 2 3 / 4 5 6 / 7 8 0

1. The first five expanded nodes are the current states listed in the first table.

2. Total generated: 500 unique states (including the initial state) by the time the goal is removed from the queue; 311 states were expanded including the goal.

3. Blank moves: Right → Up → Left → Down → Left → Down → Right → Right. The goal is reached in 8 moves.

4. BFS explores states in increasing depth. Since every move costs one, the first goal found has the fewest possible moves.